

CSCI 315 Assignment 6 Write Up

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Part 1:

Unfortunately, model was not able to correctly classify 'knish' as a noun.

```
0%|          | 0/300 [00:00<?, ?it/s]
0%|          | 1/300 [00:00<00:54, 5.53it/s]
19%|██       | 56/300 [00:00<00:00, 244.34it/s]
36%|████    | 109/300 [00:00<00:00, 353.87it/s]
53%|██████  | 158/300 [00:00<00:00, 401.30it/s]
68%|███████ | 204/300 [00:00<00:00, 419.61it/s]
83%|████████ | 249/300 [00:00<00:00, 421.78it/s]
98%|█████████| 294/300 [00:00<00:00, 421.50it/s]
100%|██████████| 300/300 [00:00<00:00, 347.94it/s]
```

Word	Predicted	Actual
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The	DET	DET
dog	NN	NN
ate	V	V
the	DET	DET
apple	NN	NN

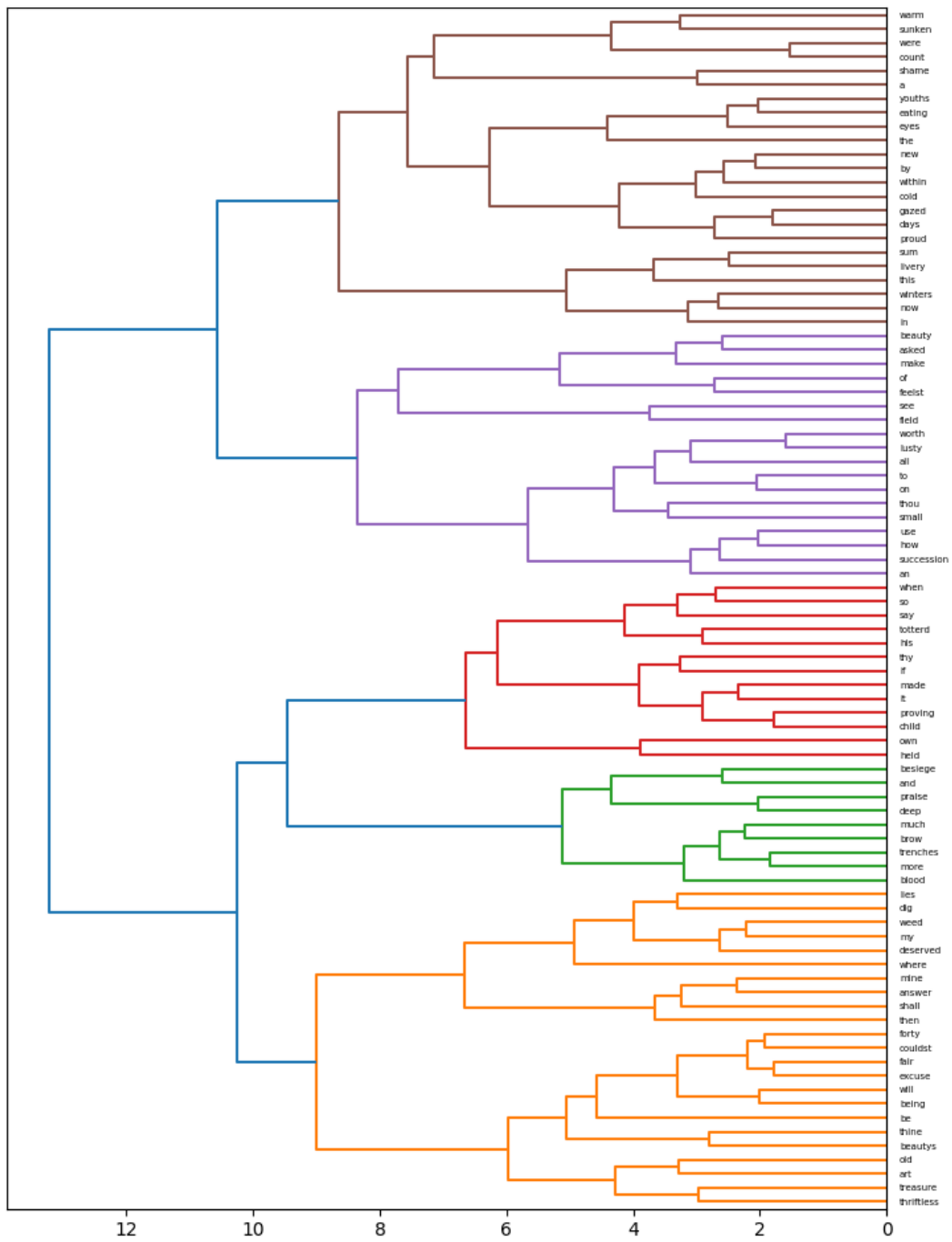
```
test("The dog ate the knish".split(), ["DET", "NN", "V", "DET", "NN"])
```

Word	Predicted	Actual
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The	DET	DET
dog	NN	NN
ate	V	V
the	DET	DET
knish	ADJ	NN

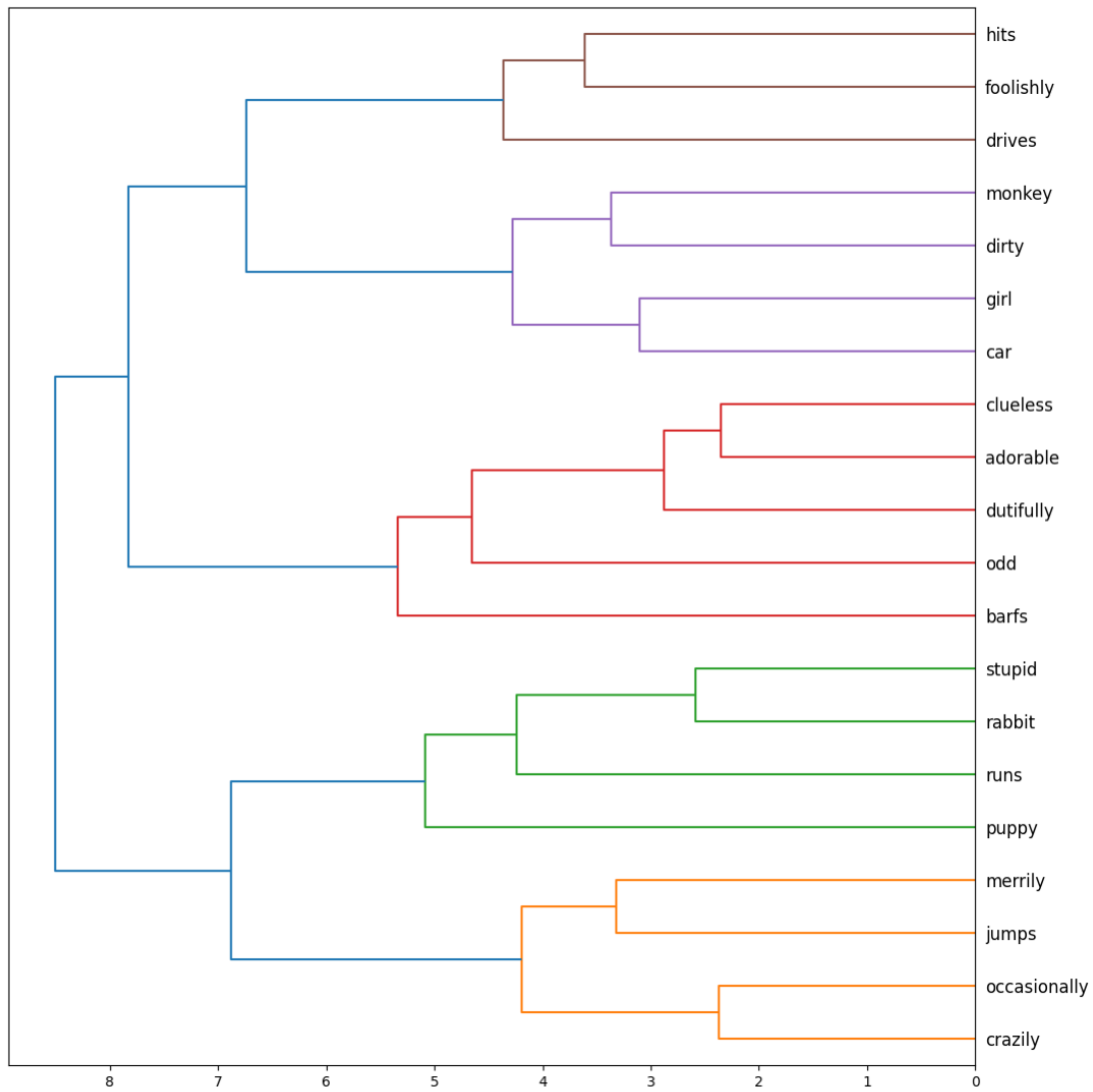
Part 2:

Dendrogram of Sonnet with 100 Epochs



Too hard to read, not even accurate

Dendrogram of Random Sentence Generation with 100 Epochs



Easier to read, but gets it right 50% of the time?

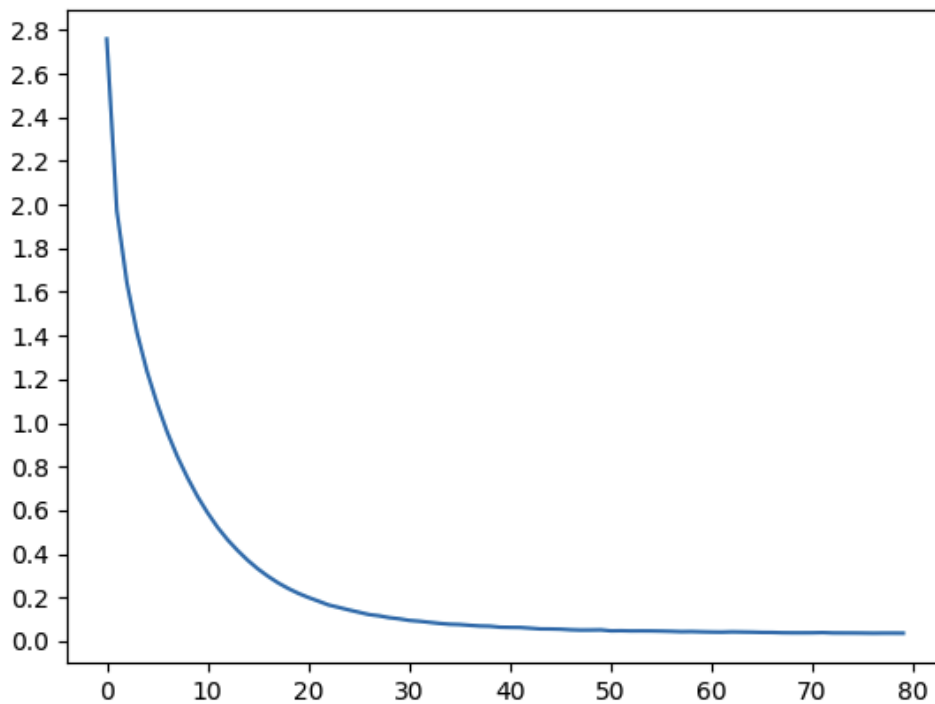
Part 3:

CPU 80 Epochs: 10 minutes and 45 seconds

GPU 80 Epochs: 1 minute and 13 seconds

- Used RTX 5090 32vCPU through RunPod (<https://www.runpod.io/>)
- Kept RunPod instance open for roughly 35 minutes to do troubleshooting with setup and incurred roughly 50 cents

Training loss over epochs for English to Spanish



Visualizing attention for English to Spanish

`evaluateAndShowAttention('se dice que tokió es una ciudad muy segura')`

`evaluateAndShowAttention('las personas a veces sus capacidades')`

`evaluateAndShowAttention('es muy raro que eso por esta personas')`

`evaluateAndShowAttention('hoy no me quiero comida china')`

